

Journal Entry - Rasterization Pipeline & WebGL II

Elina Adibi

(60%) Concepts & Application (2-3 paragraphs)

In this module, I've learned about advanced WebGL techniques and how to manage a camera within a 3D environment, which has been both fascinating and challenging. The most interesting aspect for me was implementing the camera's movement and rotation, allowing for interactive navigation through the scene. Understanding matrix transformations and their application to achieve camera control was rewarding :)

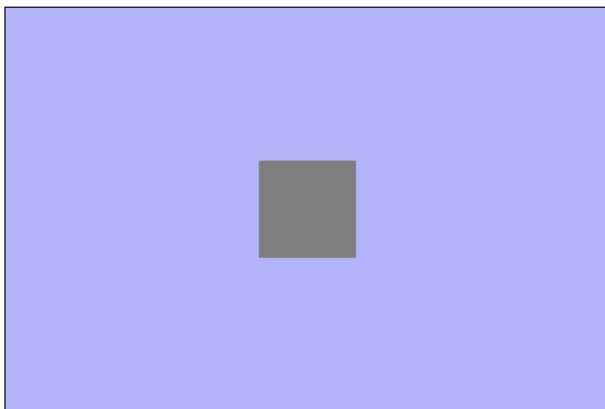
I've also learned to animate objects using time-based functions, which adds life to the scene. The oscillating color animation of the sphere using a sine wave was a way to understand uniform variables in shaders and how they affect rendering.

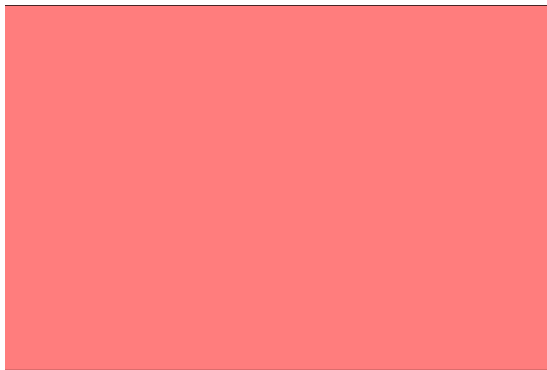
One challenge was ensuring the order of matrix multiplications to combine translations and rotations correctly, affecting the camera's final position and orientation.

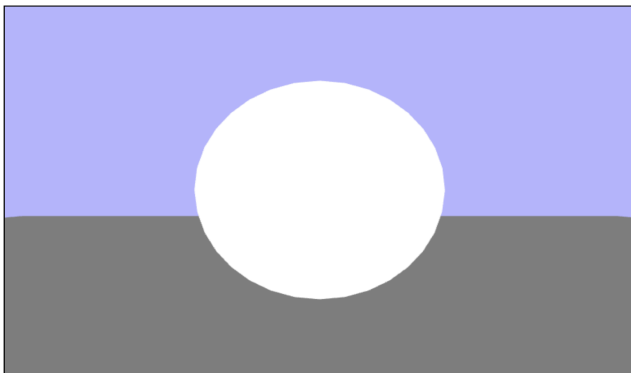
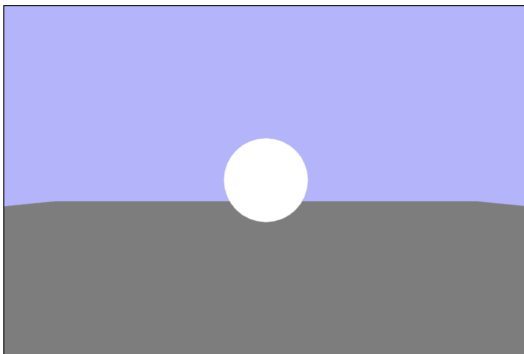
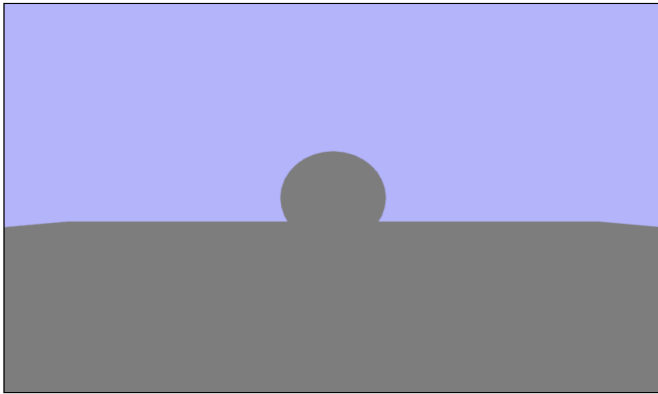
I'm left wondering about optimization techniques for such transformations and whether there's a more efficient way to handle many simultaneous object updates in a scene.

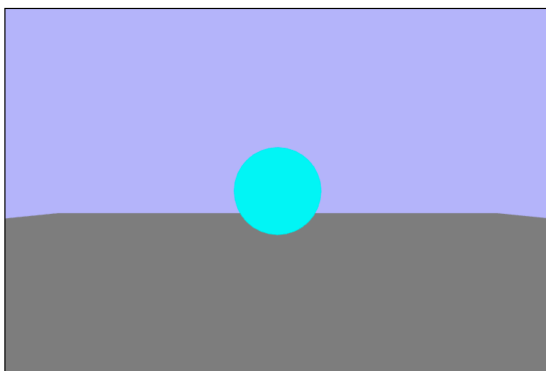
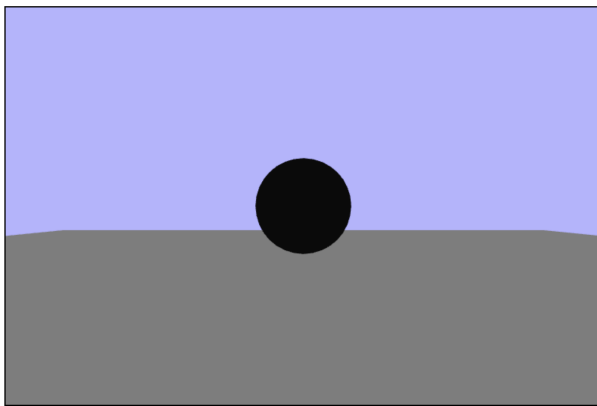
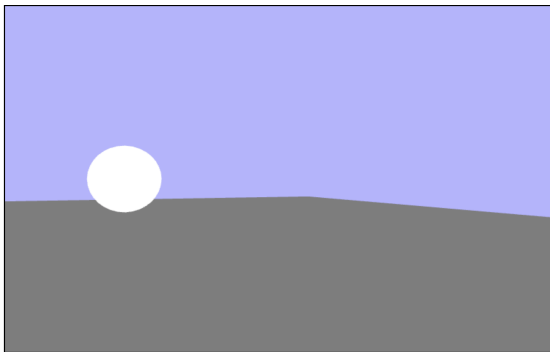
(20%) Visual Documentation

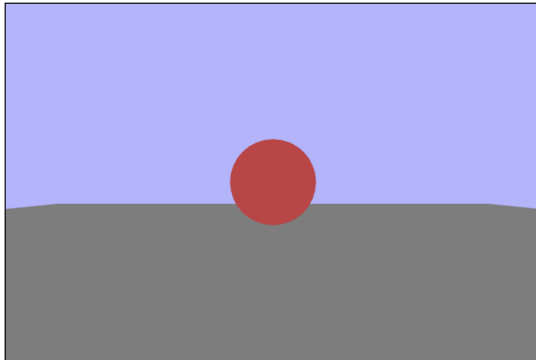
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(20%) Support Utilization

- (5%) How effectively did you and your partner work as a team?
My partner and I worked well together during the module. We talked a lot and helped each other understand this week's material and solve homework together.
- (5%) Did you attend office hours or seek help from the TA or professor?
Unfortunately, I could not attend office hours this week.
- (10%) Submit a link to your most useful shared conversation with the [course GPT](#). This conversation should demonstrate how you used AI to further your understanding of the concepts, troubleshoot issues, or explore new ideas. Do not delete this conversation or the link will break.

<https://chat.openai.com/share/fb3d06b5-ac93-4ad8-b675-e17971ae8806>

Disclaimer: While AI tools like ChatGPT can be helpful (especially ChatGPT Plus), they are not infallible and sometimes produce incorrect or misleading information (referred to as 'hallucinations'). Always cross-check information. Note that the quality of ChatGPT output is proportional to your typed input (the prompt). Prefer expansive, long prompts, over terse ones.